ARCHIT SHARMA

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EDUCATION

New York University — Sept 2024 - May 2026
M.S. Mechatronics and Robotics — *GPA: 3.8*

- Relevant Labwork: Assisted with Ultra OP1 teleoperation data collection to fine-tune a pi0.5 VLA manipulation policy.

Birla Institute of Technology Pilani - Goa — Aug 2019 - May 2023
B.Tech Mechanical Engineering — *GPA: 3.4*

SKILLS

- **Programming Languages:** Python, C/C++, C#, MATLAB
- **Robotics and Simulation:** ROS/ROS2, NVIDIA Isaac Lab, MuJoCo, RLlib, Stable Baselines 3, MINK/PINK, Docker
- **Machine Learning and Computer Vision:** PyTorch, TensorFlow, OpenCV, Open3D, PCL
- **CAD and Embedded Systems:** SolidWorks, Onshape, COMSOL, Blender, Unity, Arduino, Raspberry Pi, ESP32

PUBLICATIONS

Neural Network-Based Control for Vehicle Suspension Systems with Active Damping
Archit Sharma, Pravin M Singru — *The 30th International Congress on Sound and Vibration, 2024*

EXPERIENCE

Applied Dynamics and Optimization Laboratory — Sept 2025 - Present
Graduate Student Researcher — *NYU, New York*

- Integrated COMSOL fluid simulation with MuJoCo to enable energy analysis and trajectory optimization of bio-inspired fish (carangiform) locomotion.
- Developed a convex optimization framework to bridge the information gaps between the fluid simulation and rigid-body dynamics.

Developers' Society of BITS Goa — May 2020 - Sept 2024
3D Asset Designer — *BITS Pilani, Goa*

- Modeled, rigged, and optimized 3D assets - including character designs, concept art, and environments - in Blender for integration into Unity games.
- Organized a Unity development course for 100+ students, mentoring them through Unity editor workflows, culminating in 12 published games.

Dynamics and Vibrations Lab — Aug 2021 - Jan 2024
Undergraduate Student Researcher — *BITS Pilani, Goa*

- Implemented and benchmarked PD, PID, and LQR controllers for a quarter-car semi-active suspension system.
- Researched, implemented, and published a novel neural network-based adaptive controller for the system.

Computational Intelligence Lab — Jan 2023 - Aug 2023
Research Intern — *IISc, Bengaluru*

- Conducted musculoskeletal analysis of deep skeletal muscles using inverse kinematics (IK) and dynamics (ID) on OptiTrack motion captures for several participants.
- Developed a signal processing pipeline in MATLAB using FFT and signal filters (band-pass, Kalman) to successfully denoise and analyze IMU and EMG sensor data.

PROJECTS

Optimal Control and Reinforcement Learning — Sept 2025 - Dec 2025

- Trained a Unitree Go2 quadruped robot in NVIDIA Isaac Lab using PPO deep reinforcement learning algorithm with a custom reward model to achieve a stable gait on rough and stepped terrains.
- Implemented a nonlinear MPC enabling simulated drones to execute agile maneuvers.
- Implemented an MPPI controller for a 3R planar manipulator to achieve accurate end-effector trajectory tracking.

Robot Perception and Computer Vision — Sept 2025 - Dec 2025

- Implemented a custom optical flow algorithm in OpenCV for multi-object vehicle tracking.
- Developed a content-based image retrieval pipeline using a ResNet feature extraction model trained on the VPR dataset to accurately identify locations in New York.

Formation Control and Flocking for Drone Swarms — Mar 2025 - May 2025

- Programmed decentralized simulated drone swarms using Reynolds' Flocking theorem to implement collective behavior in Unity under safety constraints enforced through a potential field controller.
- Implemented and tested swarm optimization algorithms such as PSO for exploratory theory.

Gesture Controlled Hexapod — Nov 2024 - May 2025

- Programmed an Arduino microcontroller to drive a hexapod, interfaced with an ESP32 module to stream live video feed over Wi-Fi.
- Designed a wrist-mounted ESP32 controller to sample analog inputs from flex sensors and wirelessly transmit control states.
- Implemented gait algorithms assigned to different gestures for stable, multidirectional legged motion.